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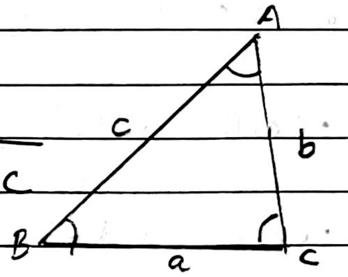
Trigonometry

• Properties of Triangle

The Sine law

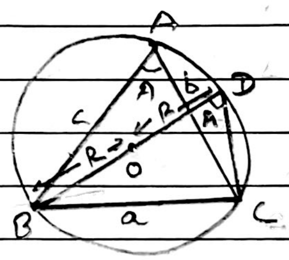
Statement :- In any triangle, the sides are proportional to the sine of their opposite angles.



i.e. In any ΔABC,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Where R is the circum radius of ΔABC



In right angled ΔBCD,

$$\sin A = \frac{BC}{BD}$$

$$\Rightarrow \sin A = \frac{a}{2R}$$

$$\Rightarrow 2R = \frac{a}{\sin A}$$

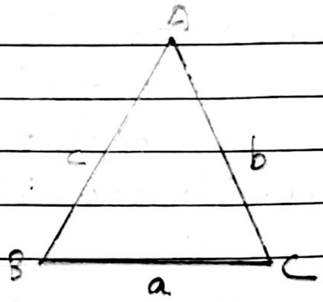
Notes: By sine Law, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

We have,

- $a = 2R \sin A$
- $b = 2R \sin B$
- $c = 2R \sin C$

- $\sin A = \dfrac{a}{2R}$
- $\sin B = \dfrac{b}{2R}$
- $\sin C = \dfrac{c}{2R}$

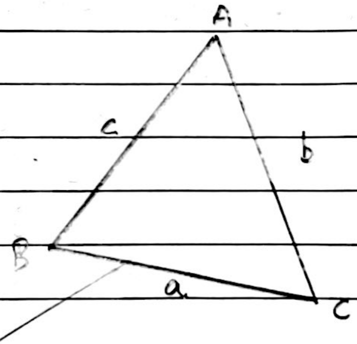
Cosine Law



Statement: In any ΔABC,

- $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$
- $\cos B = \dfrac{a^2 + c^2 - b^2}{2ac}$
- $\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$

The Projection Law



Statement: In any ΔABC

- $a = b\cos C + c\cos B$
- $b = a\cos C + c\cos A$
- $c = a\cos B + b\cos A$

* proof of projection law using sine law:

- $c = a\cos B + b\cos A$

Taking R.H.S $= a\cos B + b\cos A$

$= 2R\sin A\cos B + 2R\sin B \cdot \cos A$

$= 2R(\sin A \cdot \cos B + \cos A \cdot \sin B)$

$= 2R\sin(A+B)$

$= 2R\sin(\pi - C)$

$= 2R\sin C$

$= c$

$=$ L.H.S

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- $a = b\cos C + c\cos B$

R.H.S $= b\cos C + c\cos B$

$= 2R\sin B\cos C + 2R\sin C\cos B$

$= 2R(\sin B\cdot\cos C + \cos B\cdot\sin C)$

$= 2R\sin(B+C)$

$= 2R\sin(\pi - A)$

$= 2R\sin A$

$= a$

$=$ L.H.S

Proved.

- $b = a\cos C + c\cos A$

R.H.S $= a\cos C + c\cos A$

$= 2R\sin A\cos C + 2R\sin C\cdot\cos A$

$= 2R(\sin A\cdot\cos C + \cos A\cdot\sin C)$

$= 2R\sin(A+C)$

$= 2R\sin(\pi - B)$

$= 2R\sin B$

$= b$

$=$ L.H.S

Proved.

proof of projection using cosine Law

- $b = a\cos C + \cos A$

R.H.S $= a\cos C + c\cos A$

$$= \cancel{a} \times \frac{a^2+b^2-c^2}{2\cancel{a}b} + \cancel{c} \times \frac{b^2+c^2-a^2}{2b\cancel{c}}$$

$$= \frac{\cancel{a^2}+b^2-\cancel{c^2}+b^2+\cancel{c^2}-\cancel{a^2}}{2b}$$

$$= \frac{\cancel{2}b^{\cancel{2}}}{\cancel{2}\cancel{b}}$$

$= b =$ L.H.S

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- $a = b\cos C + c\cos B$

R.H.S $= b\cos C + c\cos B$

$$= \cancel{b} \times \frac{a^2+b^2-c^2}{2a\cancel{b}} + \cancel{c} \times \frac{a^2+c^2-b^2}{2a\cancel{c}}$$

$$= \frac{a^2+\cancel{b^2}-\cancel{c^2}+a^2+\cancel{c^2}-\cancel{b^2}}{2a}$$

$$= \frac{\cancel{2}a^2}{\cancel{2}\cancel{a}}$$

$= a =$ L.H.S

proved

- $c = a\cos B + b\cos A$

R.H.S $= a\cos B + b\cos A$

$$= \cancel{a} \times \frac{a^2+c^2-b^2}{2\cancel{a}c} + \cancel{b} \times \frac{b^2+c^2-a^2}{2\cancel{b}c}$$

$$= \frac{\cancel{a^2}+c^2-\cancel{b^2}+\cancel{b^2}+c^2-\cancel{a^2}}{2c}$$

$$= \frac{\cancel{2}c^{\cancel{2}}}{\cancel{2}\cancel{c}}$$

$= c =$ ~~R~~.H.S

Proved

The tangent Law:

In any ΔABC,

(i) $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b}\cot\frac{C}{2}$

(ii) $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c}\cot\frac{A}{2}$

(iii) $\tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a}\cot\frac{B}{2}$

Proof

R.H.S $= \frac{c-a}{c+a}\cot\frac{B}{2}$

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$$= \frac{2R\sin C - 2R\sin A}{2R\sin C + 2R\sin A} \cot\frac{B}{2}$$

$$= \frac{\sin C - \sin A}{\sin C + \sin A} \cot\frac{B}{2}$$

Formula:

- $\sin C - \sin D = 2\cos\frac{C+D}{2} \cdot \sin\frac{C-D}{2}$
- $\sin C + \sin D = 2\sin\frac{C+D}{2} \cdot \cos\frac{C-D}{2}$
- $\cot\left(\frac{\pi}{2} - \theta\right) = \tan\theta$

$$= \frac{\cancel{2}\sin\left(\frac{C-A}{2}\right) \cdot \cos\left(\frac{C+A}{2}\right)}{\cancel{2}\sin\left(\frac{C+A}{2}\right) \cdot \cos\left(\frac{C-A}{2}\right)} \times \cot\frac{B}{2}$$

$$= \cot\left(\frac{C+A}{2}\right) \cdot \tan\left(\frac{C-A}{2}\right) \cdot \cot\frac{B}{2}$$

$$= \cot\left(\frac{\pi}{2} - \frac{B}{2}\right) \cdot \tan\left(\frac{C-A}{2}\right) \cdot \cot\frac{B}{2}$$

$$= \tan\frac{B}{2} \cdot \tan\left(\frac{C-A}{2}\right) \cdot \cot\frac{B}{2}$$

$$= \cancel{\tan\frac{B}{2}} \cdot \tan\left(\frac{C-A}{2}\right) \cdot \frac{1}{\cancel{\tan B/2}}$$

$$= \tan\left(\frac{C-A}{2}\right) = L.H.S$$

Proved

- $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cot\frac{C}{2}$

$$R.H.S = \frac{a-b}{a+b} \cot\frac{C}{2}$$

$$= \frac{2R\sin A - 2R\sin B}{2R\sin A + 2R\sin B} \times \cot\frac{C}{2}$$

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$$= \frac{\sin A - \sin B}{\sin A + \sin B} \times \cot\frac{C}{2}$$

$$= \frac{\cancel{2}\cos\left(\frac{A+B}{2}\right)\cdot\sin\left(\frac{A-B}{2}\right)}{\cancel{2}\sin\left(\frac{A+B}{2}\right)\cdot\cos\left(\frac{A-B}{2}\right)} \times \cot\frac{C}{2}$$

$$= \cot\left(\frac{A+B}{2}\right)\cdot\tan\left(\frac{A-B}{2}\right) \times \cot\frac{C}{2}$$

$$= \cot\left(\frac{\pi}{2} - \frac{C}{2}\right)\cdot\tan\left(\frac{A-B}{2}\right) \times \cot\frac{C}{2}$$

$$= \tan\frac{C}{2}\cdot\tan\left(\frac{A-B}{2}\right) \times \cot\frac{C}{2}$$

$$= \tan\left(\frac{A-B}{2}\right) = L.H.S$$

Proved

- $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c} \cot\frac{A}{2}$

$$R.H.S = \frac{2R\sin B - 2R\sin C}{2R\sin B + 2R\sin C} \cot\frac{A}{2}$$

$$= \frac{\sin B - \sin C}{\sin B + \sin C} \times \cot\frac{A}{2}$$

$$= \frac{\cancel{2}\cos\left(\frac{B+C}{2}\right) \times \sin\left(\frac{B-C}{2}\right)}{\cancel{2}\sin\left(\frac{B+C}{2}\right) \times \cos\left(\frac{B-C}{2}\right)} \times \cot\frac{A}{2}$$

$$= \cot\left(\frac{B+C}{2}\right) \times \tan\left(\frac{B-C}{2}\right) \times \cot\frac{A}{2}$$

$$= \cot\left(\frac{\pi}{2} - \frac{A}{2}\right) \times \tan\left(\frac{B-C}{2}\right) \times \cot\frac{A}{2}$$

$$= \tan\frac{A}{2} \times \tan\left(\frac{B-C}{2}\right) \times \cot\frac{A}{2}$$

$$= \tan\left(\frac{B-C}{2}\right) = L.H.S$$

Proved

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The half-angle law

↳ In any $\triangle ABC$,

①

a) $\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$

b) $\sin\frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{ac}}$

c) $\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

②

a) $\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$

b) $\cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ac}}$

c) $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$

③

a) $\tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$

b) $\tan\frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$

c) $\tan\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$

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(4)

a) $\cot\frac{A}{2} = \sqrt{\frac{s(s-a)}{(s-b)(s-c)}}$

b) $\cot\frac{B}{2} = \sqrt{\frac{s(s-b)}{(s-a)(s-c)}}$

c) $\cot\frac{C}{2} = \sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$

Where $s = \frac{a+b+c}{2}$, is the semi-perimeter of triange.

Proof

$\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$

We know,

$\cos A = 1 - 2\sin^2\frac{A}{2}$

or, $2\sin^2\frac{A}{2} = 1 - \cos A$

or, $\sin\frac{A}{2} = \sqrt{\frac{1-\cos A}{2}}$

or, $\sin\frac{A}{2} = \sqrt{\frac{1-\left(\frac{b^2+c^2-a^2}{2bc}\right)}{2}}$

or, $\sin\frac{A}{2} = \sqrt{\frac{2bc-b^2-c^2+a^2}{4bc}}$

or, $\sin\frac{A}{2} = \sqrt{\frac{a^2-(b-c)^2}{4bc}}$

or, $\sin\frac{A}{2} = \sqrt{\frac{(a+b-c)(a-b+c)}{4bc}}$

or, $\sin\frac{A}{2} = \sqrt{\frac{(a+b+c-2c)(a+b+c-2b)}{4bc}}$

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or, $\sin\frac{A}{2} = \sqrt{\frac{(2s-2c)(2s-2b)}{4bc}}$ $\left[\because s = \frac{a+b+c}{2}\right]$

or, $\sin\frac{A}{2} = \sqrt{\frac{\cancel{4}(s-c)(s-b)}{\cancel{4}bc}}$

$\therefore \sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$

proved.

• $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$

Proof

We have,

$\cos C = 2\cos^2\frac{C}{2} - 1$

$\Rightarrow 2\cos^2\frac{C}{2} = \cos C + 1$

$\Rightarrow 2\cos^2\frac{C}{2} = \frac{a^2+b^2-c^2}{2ab} + 1$

$\Rightarrow 2\cos^2\frac{C}{2} = \frac{a^2+b^2-c^2+2ab}{2ab}$

$\Rightarrow 2\cos^2\frac{C}{2} = \frac{(a+b)^2-c^2}{2ab}$

$\Rightarrow \cos^2\frac{C}{2} = \frac{(a+b)^2-c^2}{4ab}$

$\Rightarrow \cos^2\frac{C}{2} = \frac{(a+b+c)(a+b-c)}{4ab}$

$\Rightarrow \cos^2\frac{C}{2} = \frac{2s\ (a+b+c-2c)}{4ab}$ $[\because 2s = a+b+c]$

$\Rightarrow \cos^2\frac{C}{2} = \frac{2s\ (2s-2c)}{4ab}$

$\Rightarrow \cos^2\frac{C}{2} = \frac{\cancel{2}\times\cancel{2}s\ (s-c)}{\cancel{4}ab}$

$\Rightarrow \cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$

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- $\sin\frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{ac}}$

We have,

$\cos B = 1 - 2\sin^2\frac{B}{2}$

$\Rightarrow 2\sin^2\frac{B}{2} = 1 - \cos B$

$\Rightarrow 2\sin^2\frac{B}{2} = 1 - \frac{a^2+c^2-b^2}{2ac}$

$\Rightarrow 2\sin^2\frac{B}{2} = \frac{2ac-a^2-c^2+b^2}{2ac}$

$\Rightarrow \sin^2\frac{B}{2} = \frac{b^2-(a-c)^2}{4ac}$

$\Rightarrow \sin^2\frac{B}{2} = \frac{(b+a-c)(b-a+c)}{4ac}$

$\Rightarrow \sin^2\frac{B}{2} = \frac{(a+b+c-2c)(a+b+c-2a)}{4ac}$

$\Rightarrow \sin^2\frac{B}{2} = \frac{(2s-2c)(2s-2a)}{4ac}$ $\left[\because s = \frac{a+b+c}{2}\right]$

$\Rightarrow \sin^2\frac{B}{2} = \frac{\cancel{4}(s-c)(s-a)}{\cancel{4}ac}$

$\Rightarrow \sin\frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ac}}$

Proved

- $\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

We have,

$\cos C = 1 - 2\sin^2\frac{C}{2}$

$\Rightarrow 2\sin^2\frac{C}{2} = 1 - \cos C$

$\Rightarrow 2\sin^2\frac{C}{2} = 1 - \frac{a^2+b^2-c^2}{2ab}$

$$\Rightarrow 2\sin^2\frac{C}{2} = \frac{2ab - a^2 + b^2 - c^2}{2ab}$$

$$\Rightarrow \sin^2\frac{C}{2} = \frac{b^2 - (a-c)^2}{4ab}$$

$$\Rightarrow \sin^2\frac{C}{2} = \frac{(b-a+c)(b+a-c)}{4ab}$$

$$\Rightarrow \sin^2\frac{C}{2} = \frac{(a+b+c-2a)\ (a+b+c-2c)}{4ab}$$

$$\Rightarrow \sin^2\frac{C}{2} = \frac{(2s-2a)\ (2s-2c)}{4ab}$$

$$\Rightarrow \sin^2\frac{C}{2} = \frac{\cancel{2}(s-a) \times \cancel{2}(s-c)}{\cancel{4}ab}$$

$$\Rightarrow \sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-c)}{ab}}$$

Proved

- $$\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$$

$\Rightarrow$ We have,

$$\cos A = 2\cos^2\frac{A}{2} - 1$$

$$\Rightarrow 2\cos^2\frac{A}{2} = \cos A + 1$$

$$\Rightarrow 2\cos^2\frac{A}{2} = \frac{b^2+c^2-a^2}{2bc} + 1$$

$$\Rightarrow 2\cos^2\frac{A}{2} = \frac{b^2+c^2-a^2+2bc}{2bc}$$

$$\Rightarrow \cos^2\frac{A}{2} = \frac{(b+c)^2 - a^2}{4bc}$$

$$\Rightarrow \cos^2\frac{A}{2} = \frac{(b+c+a)\ (b+c-a)}{4bc}$$

$$\Rightarrow \cos^2\frac{A}{2} = \frac{2s\ (a+b+c-2a)}{4bc} \quad [\because 2s = a+b+c]$$

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$\Rightarrow \cos^2 \frac{A}{2} = \frac{\cancel{2}s \times \cancel{2}(s-a)}{\cancel{4}bc}$

$\Rightarrow \cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$

proved

- $\cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ac}}$

$\Rightarrow$ we have,

$\cos B = 2\cos^2 \frac{B}{2} - 1$

$\Rightarrow 2\cos^2 \frac{B}{2} = \cos B + 1$

$\Rightarrow 2\cos^2 \frac{B}{2} = \frac{a^2+c^2-b^2}{2ac} + 1$

$\Rightarrow \cos^2 \frac{B}{2} = \frac{a^2+c^2-b^2+2ac}{4ac}$

$\Rightarrow \cos^2 \frac{B}{2} = \frac{(a+c)^2 - b^2}{4ac}$

$\Rightarrow \cos^2 \frac{B}{2} = \frac{(a+b+c)\,(a+c-b)}{4ac}$

$\Rightarrow \cos^2 \frac{B}{2} = \frac{2s\,(a+b+c-2b)}{4ac}$

$\Rightarrow \cos^2 \frac{B}{2} = \frac{2s\,(2s-2b)}{4ac}$

$\Rightarrow \cos^2 \frac{B}{2} = \frac{\cancel{2}s \times \cancel{2}(s-b)}{\cancel{4}ac}$

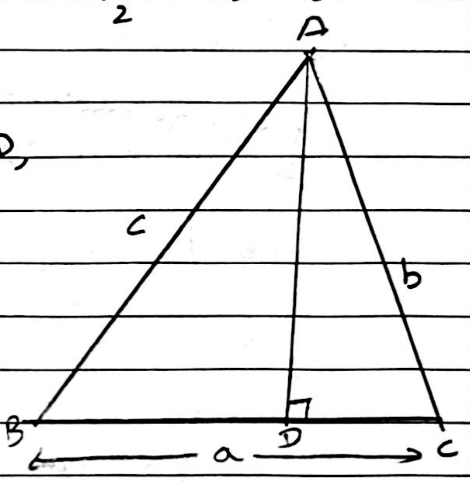
$\Rightarrow \cos^2 \frac{B}{2} = \frac{s(s-b)}{ac}$

$\Rightarrow \cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ac}}$

Area of Triangle:

→ In any ΔABC:

① $\Delta = \frac{1}{2} ab\sin C = \frac{1}{2} bc\sin A = \frac{1}{2} ca\sin B$



Proof:

In right angled ΔACD,

$\sin C = \frac{AD}{AC}$

$\sin C = \frac{AD}{b}$

or, $AD = b\sin C$

We have,

$\Delta = \frac{1}{2} \times \text{base} \times \text{height}$

$= \frac{1}{2} \times BC \times AD$

$= \frac{1}{2} \times ab\sin C$ $[\because \text{From (i)}]$

Similarly, we can prove $\Delta = \frac{1}{2} bc\sin A$ and

$\Delta = \frac{1}{2} ca\sin B$

② $\Delta = \frac{abc}{4R}$; R = circum radius

Proof:

$\Delta = \frac{1}{2} ab\sin C$

$= \frac{1}{2} ab \times \frac{c}{2R}$ $\left(\because \frac{c}{\sin C} = 2R\right)$

$\therefore \Delta = \dfrac{abc}{4R}$

(3) $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

proof

$\Delta = \dfrac{1}{2} ab \sin C$

$= \dfrac{1}{\cancel{2}} ab \times \cancel{2} \sin \dfrac{C}{2} . \cos \dfrac{C}{2}$

$= ab \sqrt{\dfrac{(s-a)(s-b)}{ab}} \times \sqrt{\dfrac{s(s-c)}{ab}}$

$= ab \sqrt{\dfrac{s(s-a)(s-b)(s-c)}{a^2 b^2}}$

$= \dfrac{\cancel{ab} \sqrt{s(s-a)(s-b)(s-c)}}{\cancel{ab}}$

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$\therefore \Delta = \sqrt{s(s-a)(s-b)(s-c)}$

(4) $\Delta = \dfrac{1}{4} \sqrt{2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4}$

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Exercise-6

① c) In ΔXYZ, if $x=3, y=5, z=6$ find Δ, R and $\tan\frac{X}{2}$.

→ Given,

In ΔXYZ, $x=3, y=5, z=6$

we have,

$$s = \frac{x+y+z}{2} = \frac{3+5+6}{2} = 7$$

$$\therefore \Delta = \sqrt{s(s-x)(s-y)(s-z)}$$
$$= \sqrt{7(7-3)(7-5)(7-6)}$$
$$= \sqrt{7\times4\times2\times1}$$
$$= \sqrt{14\times2\times2}$$
$$= 2\sqrt{14} \text{ sq.unit.}$$

Again,

$$\Delta = \frac{xyz}{4R}$$

$$\text{or, } 2\sqrt{14} = \frac{3\times5\times6}{4R}$$

$$\text{or, } R = \frac{15\times6}{8\sqrt{14}}$$

$$\therefore R = \frac{45\sqrt{14}}{56} \text{ ans.}$$

Last,

$$\tan\frac{X}{2} = \sqrt{\frac{(s-y)(s-z)}{s(s-x)}}$$
$$= \sqrt{\frac{(7-5)(7-6)}{7(7-3)}}$$
$$= \sqrt{\frac{2\times1}{7\times\cancel{4}2}}$$
$$= \frac{1}{\sqrt{14}} \text{ ans.}$$

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a) In $\triangle ABC$, if $a=3$, $b=4$ and $c=5$ prove that $\sin 2A = \frac{24}{25}$

↳ Soln,

$\cancel{\Delta}$ $s = \frac{a+b+c}{2} = \frac{3+4+5}{2} = 6$

$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{6(6-3)(6-4)(6-5)}$

$= \sqrt{6 \times 3 \times 2 \times 1}$

$= 6$

Now,

$\Delta = \frac{abc}{4R}$

$6 = \frac{3 \times \cancel{4} \times 5}{\cancel{4}R} \Rightarrow R = \frac{15}{6}$

$\sin A = \frac{a}{2R} \Rightarrow \frac{3}{\frac{\cancel{2} \times 15}{\cancel{6}3}} \Rightarrow \frac{3}{1} \times \frac{3}{15} = \frac{9}{15}$

$\cos A = \frac{b^2+c^2-a^2}{2bc}$

$= \frac{4^2+5^2-3^2}{2 \times 4 \times 5}$

$= \frac{16+25-9}{40} = \frac{4}{5}$

Then,

$\sin 2A = 2 \sin A . \cos A$

$= 2 \times \frac{9}{15} \times \frac{4}{5}$

$= \frac{24}{25}$

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2. In any ΔABC prove that.

d) $\frac{b^2+c^2-a^2}{4\cot A} = \Delta$

$L.H.S = \frac{b^2+c^2-a^2}{4\cot A}$

$= \frac{b^2+c^2-a^2}{2bc} \times \frac{2bc}{4\cot A}$

$= \cos A \times \frac{1}{2} bc \times \tan A$

$= \cancel{\cos A} \times \frac{1}{2} \times bc \times \frac{\sin A}{\cancel{\cos A}}$

$= \frac{1}{2} bc \sin A$

$= \Delta = R.H.S$

Proved.

e) $\frac{\cos A}{a} + \frac{a}{bc} = \frac{\cos B}{b} + \frac{b}{ca} = \frac{\cos C}{c} + \frac{c}{ab}$

$\hookrightarrow$ $L.H.S = \frac{\cos A}{a} + \frac{a}{bc}$

$= \frac{bc\cos A + a^2}{abc}$

$= \frac{\cancel{bc} \times \frac{b^2+c^2-a^2}{2\cancel{bc}} + a^2}{abc}$

$= \frac{b^2+c^2-a^2+2a^2}{2abc}$

$= \frac{a^2+b^2+c^2}{2abc}$

$M.S = \frac{\cos B}{b} + \frac{b}{ca}$

$= \frac{ca\cos B + b^2}{abc}$

Formula :- • $\sin(A+B) \times \sin(A-B)$
$= \sin^2 A - \sin^2 B$

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$$= \frac{\cancel{ca}\,\frac{a^2+c^2-b^2}{2\cancel{ac}} + b^2}{abc}$$

$$= \frac{a^2+c^2+b^2}{2abc}$$

$$\text{R.H.S} = \frac{\cos C}{c} + \frac{c}{ab}$$

$$= \frac{ab\cos C + c^2}{abc}$$

$$= \frac{\cancel{ab}\,\frac{a^2+b^2-c^2}{2\cancel{ab}} + c^2}{abc}$$

$$= \frac{a^2+b^2-c^2+2c^2}{2abc}$$

$$= \frac{a^2+b^2+c^2}{2abc}$$

~~Since~~, $\text{L.H.S} = \text{M.S} = \text{R.H.S} = \dfrac{a^2+b^2+c^2}{2abc}$

Proved

③ In any ΔABC, prove the followings :

b) $\dfrac{a\sin(B-C)}{b^2-c^2} = \dfrac{b\sin(C-A)}{c^2-a^2} = \dfrac{c\sin(A-B)}{a^2-b^2}$

↳ $\text{L.H.S} = \dfrac{a\sin(B-C)}{b^2-c^2}$

$$= \frac{\cancel{a}\,2R\sin A\,\sin(B-C)}{(2R\sin B)^2 - (2R\sin C)^2}$$

$$= \frac{\cancel{2R}\sin A \cdot \sin(B-C)}{2\cancel{4}R^{\cancel{2}}(\sin^2 B - \sin^2 C)}$$

$$= \frac{\sin A \cdot \cancel{\sin(B-C)}}{2R\{\sin(B+C)\ \cancel{\sin(B-C)}\}}$$

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$$= \frac{\sin A}{2R\sin(B+C)}$$

$$= \frac{\sin A}{2R(\pi - A)}$$

$$= \frac{\cancel{\sin A}}{2R\cancel{\sin A}}$$

$$= \frac{1}{2R}$$

$$M.S = \frac{b\sin(C-A)}{c^2 - a^2}$$

$$= \frac{2R\sin B \ \sin(C-A)}{(2R\sin C)^2 - (2R\sin A)^2}$$

$$= \frac{\cancel{2R}\sin B \cdot \sin(C-A)}{2\cancel{4}R^{\cancel{2}}(\sin^2 C - \sin^2 A)}$$

$$= \frac{\cancel{2}\sin B \cdot \cancel{\sin(C-A)}}{2R\{\sin(C+A)\cdot\cancel{\sin(C-A)}\}}$$

$$= \frac{\sin B}{2R\sin(\pi - B)}$$

$$= \frac{\cancel{\sin B}}{2R\cancel{\sin B}}$$

$$= \frac{1}{2R}$$

$$R.H.S = \frac{c\sin(A-B)}{a^2 - b^2}$$

$$= \frac{2R\sin C \ \sin(A-B)}{(2R\sin A)^2 - (2R\sin B)^2}$$

$$= \frac{\cancel{2R}\sin C \ \sin(A-B)}{2\cancel{4}R^{\cancel{2}}(\sin^2 A - \sin^2 B)}$$

$$= \frac{\sin C \ \cancel{\sin(A-B)}}{2R\{\sin(A+B)\cancel{(\sin(A-B))}\}}$$

$$= \frac{\sin C}{2R\sin(\pi - C)}$$

$$= \frac{\cancel{\sin C}}{2R \cancel{\sin C}}$$

$$= \frac{1}{2R}$$

$\therefore$ L.H.S = M.S = R.H.S = $\frac{1}{2R}$

Proved.

1. b) In ΔPQR, if $p=13$, $q=14$, $r=15$, find s, Δ, $\sin\frac{P}{2}$, $\cos\frac{P}{2}$, $\tan\frac{P}{2}$.

↳ Soln,

$$s = \frac{p+q+r}{2} = \frac{13+14+15}{2} = 21$$

$$\Delta = \sqrt{s(s-p)(s-q)(s-r)}$$
$$= \sqrt{21 \times (21-13)(21-14)(21-15)}$$
$$= \sqrt{21 \times 8 \times 7 \times 6}$$
$$= \sqrt{7056}$$
$$= 84 \text{ sq unit.}$$

$$\sin\frac{P}{2} = \sqrt{\frac{(s-q)(s-r)}{qr}}$$
$$= \sqrt{\frac{(21-14)(21-15)}{14 \times 15}} = \sqrt{\frac{7 \times 6}{14 \times 15}}$$
$$= \frac{1}{\sqrt{5}}$$

$$\cos\frac{P}{2} = \sqrt{\frac{s(s-p)}{qr}} = \sqrt{\frac{21(21-13)}{14 \times 15}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$$

~~Again~~ Now,

$$\tan\frac{P}{2} = \frac{\sin P/2}{\cos P/2}$$

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$$= \frac{\frac{1}{\sqrt{3}}}{\frac{\sqrt{4}}{\sqrt{3}}}$$

$$= \frac{1}{\sqrt{4}} = \frac{1}{2}$$

② a) $(b+c)\cos A + (c+a)\cos B + (a+b)\cos C = a+b+c$

↳ L.H.S $= (b+c)\cos A + (c+a)\cos B + (a+b)\cos C$

$= b\cos A + c\cos A + c\cos B + a\cos B + a\cos C + b\cos C$

$= b\cos A + a\cos B + c\cos A + a\cos C + c\cos B + b\cos C$

$= c + b + a$ [From projection Law]

$= a+b+c = $ R.H.S

b) $ac\cos B - bc\cos A = a^2 - b^2$

↳ L.H.S $= ac\cos B - bc\cos A$

$$= \cancel{ac} \times \frac{a^2+c^2-b^2}{2\cancel{ac}} - \cancel{bc} \times \frac{b^2+c^2-a^2}{2\cancel{bc}}$$

$$= \frac{a^2+\cancel{c^2}-b^2-b^2-\cancel{c^2}+a^2}{2}$$

$$= \frac{2a^2-2b^2}{2}$$

$= a^2 - b^2 = $ R.H.S

Proved.

c) $\dfrac{c - b\cos A}{b - c\cos A} = \dfrac{\cos B}{\cos C}$

↳ L.H.S $= \dfrac{c - b\cos A}{b - c\cos A}$

$$= \frac{a\cos B + \cancel{b\cos A} - \cancel{b\cos A}}{a\cos C + \cancel{c\cos A} - \cancel{c\cos A}}$$

$$= \frac{\cancel{a}\cos B}{\cancel{a}\cos C} = \frac{\cos B}{\cos C} = \text{R.H.S}$$

Proved.

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~~(3) a) $\frac{b-c}{a} \cos\frac{A}{2} = \sin\frac{B-C}{2}$~~

~~L.H.S $= \frac{b-c}{a} \cdot \cos\frac{A}{2}$~~

~~$= \frac{2R\sin B - 2R\sin C}{2R\sin A} \cdot \cos\frac{A}{2}$~~

~~$= \frac{2R(\sin(B-C))}{2R\sin A}$~~

~~$= \frac{2\sin\left(\frac{B-C}{2}\right) \cdot \cos\left(\frac{B-C}{2}\right)}{2 \times \sin A/2 \cos A/2} \times \cos\frac{A}{2}$~~

~~$= \frac{\sin\left(\frac{B-C}{2}\right) \cdot \cos\left(\frac{B-C}{2}\right)}{\sin\frac{A}{2}}$~~

3. a) $\frac{b-c}{a} \cdot \cos\frac{A}{2} = \sin\frac{B-C}{2}$

L.H.S $= \frac{b-c}{a} \cdot \cos\frac{A}{2}$

$= \frac{2R\sin B - 2R\sin C}{2R\sin A} \cdot \cos\frac{A}{2}$

$= \frac{\cancel{2R}(\sin B - \sin C)}{\cancel{2R}\sin A} \cdot \cos\frac{A}{2}$

$= \frac{\cancel{2}\cos\left(\frac{B+C}{2}\right) \cdot \sin\left(\frac{B-C}{2}\right)}{\cancel{2}\sin\left(\frac{A}{2}\right) \cdot \cancel{\cos\frac{A}{2}}} \times \cancel{\cos\frac{A}{2}}$

$= \frac{\cos\left(\frac{\pi}{2} - \frac{A}{2}\right) \cdot \sin\left(\frac{B-C}{2}\right)}{\sin\frac{A}{2}}$

$= \frac{\cancel{\sin A/2} \cdot \sin\left(B-C/2\right)}{\cancel{\sin A/2}}$

$= \sin\left(\frac{B-C}{2}\right)$ #

$\sin(A+B) = \sin(\pi - C) = \sin C$

$\cos(A+B) = \cos(\pi - C) = -\cos C$

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f) $a\cos A + b\cos B + c\cos C = 4R\sin A \cdot \sin B \sin C$

$\hookrightarrow$ L.H.S $= a\cos A + b\cos B + c\cos C$

$= 2R\sin A\cos A + 2R\sin B \cdot \cos B + 2R\sin C \cdot \cos C$

$= R(\sin 2A + \sin 2B + \sin 2C)$

$= R\left(2\sin\frac{\cancel{2}(A+B)}{\cancel{2}} \cdot \cos\frac{\cancel{2}(A-B)}{\cancel{2}}\right) + \sin 2C$

$= 2R\sin(A+B) \cdot \cos(A-B) + R\sin 2C$

$= R\left[2\sin(\pi - C) \cdot \cos(A-B) + \sin 2C\right]$

$= R[2\sin C \cdot \cos(A-B) + 2\sin C \cdot \cos C]$

$= 2R\sin C[\cos(A-B) + [-\cos(A+B)]]$

$= 2R\sin C[\cos(A-B) - \cos(A+B)]$

$= 2R\sin C \cdot 2\sin A \cdot \sin B$

$= 4R\sin A \cdot \sin B \cdot \sin C$

$=$ R.H.S

proved

h) $\frac{a^2\sin(B-C)}{\sin A} + \frac{b^2\sin(C-A)}{\sin B} + \frac{c^2\sin(A-B)}{\sin C} = 0$

$\hookrightarrow$ L.H.S

$\frac{a^2\sin(B-C)}{\sin A} + \frac{b^2\sin(C-A)}{\sin B} + \frac{c^2\sin(A-B)}{\sin C}$

$= \frac{4R^2\sin^{\cancel{2}}A \cdot \sin(B-C)}{\cancel{\sin A}} + \frac{4R^2\sin^{\cancel{2}}B\sin(C-A)}{\cancel{\sin B}}$

$+ \frac{4R^2\sin^{\cancel{2}}C\sin(A-B)}{\cancel{\sin C}}$

$= 4R^2\sin(B+C) \cdot \sin(B-C) + 4R^2\sin(A+C)$

$+ \sin(C-A) + 4R^2\sin(A+B) \cdot \sin(A-B)$

$= 4R^2[\cancel{\sin^2 B} - \cancel{\sin^2 C} + \cancel{\sin^2 C} - \cancel{\sin^2 A} + \cancel{\sin^2 A}$

$- \cancel{\sin^2 B}]$

$= 4R^2 \times 0$

$= 0 =$ R.H.S

proved

i) $\frac{b^2-c^2}{a^2}\sin 2A + \frac{c^2-a^2}{b^2}\sin 2B + \frac{a^2-b^2}{c^2}\sin 2C = 0$

→ L.H.S

$\frac{b^2-c^2}{a^2}\sin 2A + \frac{c^2-a^2}{b^2}\sin 2B + \frac{a^2-b^2}{c^2}\sin 2C$

$= \frac{b^2-c^2}{a^2}\cdot 2\sin A\cdot\cos A + \frac{c^2-a^2}{b^2}\times 2\sin B\cdot\cos B +$

$\frac{a^2-b^2}{c^2}\times 2\sin C\cdot\cos C$

$= \frac{b^2-c^2}{a^{\cancel{2}}}\times\cancel{2}\times\frac{\cancel{a}}{\cancel{2}R}\times\frac{b^2+c^2-a^2}{2bc} + \frac{c^2-a^2}{b^{\cancel{2}}}\times\cancel{2}\times\frac{\cancel{b}}{\cancel{2}R}\times$

$\frac{a^2+c^2-b^2}{2ac} + \frac{a^2-b^2}{c^{\cancel{2}}}\times\cancel{2}\times\frac{\cancel{c}}{\cancel{2}R}\times\frac{a^2+b^2-c^2}{2ab}$

$= \frac{b^4+\cancel{b^2c^2}-a^2b^2-\cancel{b^2c^2}-c^4+a^2c^2}{2Rabc} + \frac{\cancel{a^2c^2}+c^4-b^2c^2}{2Rabc}$

$\frac{-a^4-\cancel{a^2c^2}+a^2b^2}{} + \frac{a^4+\cancel{a^2b^2}-a^2c^2-\cancel{a^2b^2}-b^4+b^2c^2}{2Rabc}$

$= \frac{\cancel{b^4}-\cancel{a^2b^2}-\cancel{c^4}+\cancel{a^2c^2}+\cancel{c^4}-\cancel{b^2c^2}-\cancel{a^4}+\cancel{a^2b^2}+\cancel{a^4}-\cancel{a^2c^2}}{2Rabc}$

$\frac{\cancel{-b^4}+\cancel{b^2c^2}}{1}$

$= \frac{0}{2Rabc}$

$= 0 = R.H.S$

Proved

c) $2a\sin\frac{B}{2}\cdot\sin\frac{C}{2} = (b+c-a)\sin\frac{A}{2}$

→ L.H.S.

$2a\sin\frac{B}{2}\cdot\sin\frac{C}{2}$

$= 2a\sqrt{\frac{(s-a)(s-c)}{ac}}\times\sqrt{\frac{(s-a)(s-b)}{ab}}$

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$$= 2a\sqrt{\frac{(s-a)^2(s-b)(s-c)}{a^2bc}}$$

$$= \frac{2\cancel{a}(s-a)}{\cancel{a}}\sqrt{\frac{(s-b)(s-c)}{bc}}$$

$$= 2\left(\frac{a+b+c}{2} - a\right) \times \sin\frac{A}{2}$$

$$= (a+b+c-2a) \times \sin\frac{A}{2}$$

$$= (b+c-a)\sin\frac{A}{2} = R.H.S$$

proved

d) $\dfrac{\sin(B-C)}{\sin(B+C)} = \dfrac{b^2-c^2}{a^2}$

↳ $R.H.S = \dfrac{b^2-c^2}{a^2}$

$$= \frac{(2R\sin B)^2 - (2R\sin C)^2}{(2R\sin A)^2}$$

$$= \frac{\cancel{4R^2}(\sin^2 B - \sin^2 C)}{\cancel{4R^2}\sin^2 A}$$

$$= \frac{\sin(B+C)\cdot\sin(B-C)}{\sin(\pi-A)\cdot\sin(\pi-A)}$$

$$= \frac{\cancel{\sin(B+C)}\cdot\sin(B-C)}{\cancel{\sin(B+C)}\cdot\sin(B+C)}$$

$$= \frac{\sin(B-C)}{\sin(B+C)} = L.H.S$$

Proved

e) $\dfrac{\sin(A+B)}{\sin(A-B)} = \dfrac{c^2}{a^2-b^2}$

↳ $R.H.S = \dfrac{c^2}{a^2-b^2}$

$$= \frac{(2R\sin C)^2}{(2R\sin A)^2 - (2R\sin B)^2}$$

$$= \frac{4R^2\sin^2 C}{4R^2\sin^2 A - 4R^2\sin^2 B}$$

$$= \frac{\cancel{4R^2}(\sin^2 C)}{\cancel{4R^2}(\sin^2 A - \sin^2 B)}$$

$$= \frac{\sin C \cdot \sin C}{\sin(A+B) \cdot \sin(A-B)}$$

$$= \frac{\cancel{\sin(\pi - C)} \cdot \sin(\pi - C)}{\cancel{\sin(\pi - C)} \cdot \sin(A-B)}$$

$$= \frac{\sin(B+A)}{\sin(A-B)}$$

$= R.H.S$

proved.

g) $a\sin(B-C) + b\sin(C-A) + c\sin(A-B) = 0$

↳ L.H.S

$a\sin(B-C) + b\sin(C-A) + c\sin(A-B)$

$= 2R\sin A \cdot \sin(B-C) + 2R\sin B\sin(C-A) + 2R\sin C \cdot \sin(A-B)$

$= 2R[\sin(B+C) \cdot \sin(B-C) + \sin(C+A) \cdot \sin(C-A) + \sin(A+B) \cdot \sin(A-B)]$

$= 2R[\cancel{\sin^2 B} - \cancel{\sin^2 C} + \cancel{\sin^2 C} - \cancel{\sin^2 A} + \cancel{\sin^2 A} - \cancel{\sin^2 B}]$

$= 2R \times 0$

$= 0 = R.H.S$

Proved.

4. In any triangle, prove that:

a) $a\cos^2\frac{B}{2} + b\cos^2\frac{A}{2} = s$

$L.H.S = a\cos^2\frac{B}{2} + b\cos^2\frac{A}{2}$

$$= \cancel{a}\,\frac{(s-b)s}{\cancel{a}c} + \cancel{b} \times \frac{s(s-a)}{\cancel{b}c}$$

$$= \frac{s^2 - bs + s^2 - as}{c}$$

$$= \frac{2s^2 - s(a+b)}{c}$$

$$= s\left(\frac{2s-a-b}{c}\right)$$

$$= s\left(\frac{\cancel{a}+\cancel{b}+c-\cancel{a}-\cancel{b}}{c}\right)$$

$$= s \times \frac{\cancel{c}}{\cancel{c}}$$

$$= s = R.H.S$$

proved

b) $2\left[p\sin^2\frac{R}{2} + r\sin^2\frac{P}{2}\right] = p-q+r$

$\rightarrow$ $L.H.S = 2\left[p\sin^2\frac{R}{2} + r\sin^2\frac{P}{2}\right]$

$$= 2\left[\cancel{p}\,\frac{(s-p)(s-q)}{\cancel{p}q} + \cancel{r}\times\frac{(s-q)(s-r)}{q\cancel{r}}\right]$$

$$= 2\left[\frac{(s-p)(s-q)+(s-q)(s-r)}{q}\right]$$

$$\cancel{= 2\left[\frac{s^2-qs-ps+pq+s^2-qs-rs+sq}{q}\right]}$$

$$\cancel{= 2\left[\frac{2s^2-2qs-ps+pq-rs+sq}{q}\right]}$$

$$= \frac{2(s-q)\ (s-p+s-r)}{q}$$

$$= \frac{(2s-2q)\ (2s-p-r)}{q}$$

$$= \frac{\left[\cancel{2}\left(\frac{p+q+r}{\cancel{2}}\right)-2q\right]\left[\cancel{2}\left(\frac{p+q+r}{\cancel{2}}\right)-p-r\right]}{q}$$

$$= \frac{(p-q+r)\ (\cancel{p}+q+\cancel{r}-\cancel{p}-\cancel{r})}{q}$$

$$= \frac{\cancel{q}(p-q+r)}{\cancel{q}}$$

$$= p-q+r = R.H.S$$

proved

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g) $\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c} - \frac{1}{s} = \frac{4R}{\Delta}$

→ L.H.S

$\frac{s-b+s-a}{(s-a)(s-b)} + \frac{s-(s-c)}{s(s-c)}$

$= \frac{2s-b-a}{(s-a)(s-b)} + \frac{c}{s(s-c)}$

$= \frac{c}{(s-a)(s-b)} + \frac{c}{s(s-c)}$

$= \frac{c}{ab}\left[\frac{1}{\frac{(s-a)(s-b)}{ab}} + \frac{1}{\frac{s(s-c)}{ab}}\right]$

$= \frac{c}{ab}\left[\frac{1}{\sin^2 c/2} + \frac{1}{\cos^2 c/2}\right]$

$= \frac{c}{ab}\left[\frac{1}{\sin^2 c/2 * \cos^2 c/2}\right]$

$= \frac{c}{ab}\left[\frac{4}{(2\sin c/2 \cdot \cos c/2)^2}\right]$

$= \frac{c}{ab}\left[\frac{4}{\sin^2 c}\right]$

$= \frac{2R\cancel{\sin c}}{ab}\left[\frac{4}{\sin^{\cancel{2}} c}\right]$

$= \frac{8R}{ab\sin c}$

$= \frac{\frac{1}{2}8R}{\frac{1}{2}ab\sin c}$

$= \frac{4R}{\Delta}$ = R.H.S

Proved

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c) $b\cos^2\frac{C}{2} + c\cos^2\frac{B}{2} = \frac{1}{2}(a+b+c)$

↳ L.H.S $= b\cos^2\frac{C}{2} + c\cos^2\frac{B}{2}$

$= \cancel{b} \times \frac{s(s-c)}{a\cancel{b}} + \cancel{c} \times \frac{s(s-b)}{a\cancel{c}}$

$= \frac{s(s-c)+s(s-b)}{a}$

$= \frac{s(s-c+s-b)}{a}$

$= \frac{s(2s-b-c)}{a}$

$= s \times \frac{\cancel{a}}{\cancel{a}}$

$= \frac{a+b+c}{2}$ $\left[\because s = \frac{a+b+c}{2}\right]$

$= R.H.S$

Proved

~~b)~~

h) $(y+z-x)\left(\cot\frac{y}{2} + \cot\frac{z}{2}\right) = 2x\cot\frac{x}{2}$

↳ L.H.S $= (y+z-x)\left(\cot\frac{y}{2} + \cot\frac{z}{2}\right)$

$= (y+z-x)\left[\sqrt{\frac{s(s-y)}{(s-x)(s-z)}} + \sqrt{\frac{s(s-z)}{(s-x)(s-y)}}\right]$

$= (y+z-x)\left[\sqrt{\frac{s}{(s-x)}}\right]\left[\sqrt{\frac{s-y}{s-z}} + \sqrt{\frac{s-z}{\sqrt{s-y}}}\right]$

$= (y+z-x)\sqrt{\frac{s}{(s-x)}}\left[\frac{s-y+s-z}{\sqrt{(s-y)(s-z)}}\right]$

$= (y+z-x)\sqrt{\frac{s}{s-x}}\left[\frac{2s-y-z}{\sqrt{s-y)(s-z)}}\right]$

$= (y+z-x)\sqrt{\frac{s}{s-x}}\left[\frac{x}{\sqrt{(s-y)(s-z)}}\right]$

$$= (2s-2x) \times \frac{\sqrt{s}}{\sqrt{s-x}} \left[\frac{x}{\sqrt{s-z}\ \sqrt{s-y}} \right]$$

$$= 2\left(\sqrt{s-x}\right)^{\cancel{2}} \times \frac{\sqrt{s}}{\cancel{\sqrt{s-x}}} \left[\frac{x}{\sqrt{s-z}\ \sqrt{s-y}} \right]$$

$$= 2\sqrt{s-x} \times \sqrt{s} \left[\frac{x}{\sqrt{s-z}\ \sqrt{s-y}} \right]$$

$$= 2x \sqrt{\frac{s(s-x)}{(s-z)(s-y)}}$$

$$\cancel{= 2x \quad s(s-x)}$$

$$= 2x \cot \frac{X}{2} = R.H.S$$

Proved

d) $1 - \tan \frac{A}{2} \cdot \tan \frac{B}{2} = \frac{2c}{a+b+c}$

→ $L.H.S = 1 - \tan \frac{A}{2} \cdot \tan \frac{B}{2}$

$$= 1 - \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \times \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$$

$$= 1 - \sqrt{\frac{\cancel{(s-b)}(s-c)\cancel{(s-a)}(s-c)}{s^2 \cancel{(s-a)} \cancel{(s-b)}}}$$

$$= 1 - \frac{(s-c)}{s}$$

$$= \frac{\cancel{s} - \cancel{s} + c}{s}$$

$$= \frac{c}{\frac{a+b+c}{2}}$$

$$= \frac{2c}{a+b+c} = R.H.S$$

Proved

e) $\frac{a+b-c}{a+b+c} = \tan\frac{A}{2} \cdot \tan\frac{B}{2}$

↳ R.H.S $= \tan\frac{A}{2} \cdot \tan\frac{B}{2}$

$= \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \times \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$

$= \sqrt{\frac{\cancel{(s-b)}(s-c)\cancel{(s-a)}(s-c)}{s^2\cancel{(s-a)}\cancel{(s-b)}}}$

$= \frac{s-c}{s}$

$= \frac{\frac{a+b+c}{2} - c}{\frac{a+b+c}{2}}$

$= \frac{a+b+c-2c}{a+b+c}$

$= \frac{a+b-c}{a+b+c} =$ L.H.S

Proved.

f) $\tan^2\frac{A}{2} \cdot \tan^2\frac{B}{2} \cdot \tan^2\frac{C}{2} = \left(\frac{s-a}{s}\right)\left(\frac{s-b}{s}\right)\left(\frac{s-c}{c}\right)$

↳ L.H.S

$= \tan^2\frac{A}{2} \cdot \tan^2\frac{B}{2} \cdot \tan^2\frac{C}{2}$

$= \frac{\cancel{(s-b)}(s-c)}{s\cancel{(s-a)}} \times \frac{\cancel{(s-a)}\cancel{(s-c)}}{s\cancel{(s-b)}} \times \frac{(s-a)(s-b)}{s\cancel{(s-c)}}$

$= \frac{s-c}{s} \times \frac{s-a}{s} \times \frac{s-b}{s}$

$= \left(\frac{s-a}{s}\right) \cdot \left(\frac{s-b}{s}\right)\left(\frac{s-c}{s}\right) =$ R.H.S

Proved.

* Angle → cosine law

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8) If $a^4+b^4+c^4-2c^2(a^2+b^2)+a^2b^2=0$, Show that $\angle C = 60°$ or $120°$.

↳ Given,

$a^4+b^4+c^4-2c^2(a^2+b^2)+a^2b^2=0$

or, $(a^2)^2+(b^2)^2+(c^2)^2-2c^2(a^2+b^2)+a^2b^2=0$

or, $(a^2+b^2)^2-2a^2b^2+(c^2)^2-2c^2(a^2+b^2)+a^2b^2=0$

or, $(a^2+b^2)^2-a^2b^2+(c^2)^2-2c^2(a^2+b^2)\cancel{+a^2b^2}=0$

or, $(a^2+b^2)^2-2c^2(a^2+b^2)-a^2b^2+(c^2)^2=0$

~~or, (a^2+b^2) (a^2+b^2)~~

or, $(a^2+b^2)^2-2c^2(a^2+b^2)^2+(c^2)^2-a^2b^2=0$

or, $(a^2+b^2-c^2)^2=a^2b^2$

or, $a^2+b^2-c^2=\pm ab$

or, $\frac{a^2+b^2-c^2}{2ab}=\pm\frac{1}{2}$

or, $\cos C=\pm\frac{1}{2}$

Taking +ve,
$\cos C=\cos 60$
$\therefore C=60°$

Taking −ve
$\cos C=\cos 120°$
$\therefore C=120°$

7) If $A=2B$, then prove that either $c=b$ or $a^2=b(c+b)$.

↳ Given,

$A=2B \rightarrow (1)$

To prove: $c=b$ or $a^2=b(c+b)$

Taking sine on both sides in (1)

$\sin A=\sin 2B$

or, $\sin A=2\sin B.\cos B$

or, $\frac{a}{\cancel{2R}}=\cancel{2}\times\frac{b}{\cancel{2R}}.\frac{a^2+c^2-b^2}{\cancel{2}ac}$

or, $a^2c=b(a^2+c^2-b^2)$

or, $a^2c=a^2b+bc^2-b^3$

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or, $a^2c - a^2b = bc^2 - b^3$

or, $a^2(c-b) = b(c^2-b^2)$

~~or, $a^2(c+b) = b(c+b)(c+b)$~~

~~$\therefore a^2 = b(c+b)$~~

~~Proved~~

or, $a^2(c-b) - b(c+b)(c-b) = 0$

or, $(c-b)(a^2 - b(c+b)) = 0$

Either,

$c-b=0 \qquad a^2 - b(c+b) = 0$

$c = b \qquad a^2 = b(c+b)$

$\therefore$ Either $b=c$ or $a^2 = b(b+c)$.

10) If $\dfrac{\sin(A-B)}{\sin(A+B)} = \dfrac{a^2-b^2}{a^2+b^2}$, prove that the $\triangle ABC$ is either isosceles or right angled.

→ Given,

$$\frac{\sin(A-B)}{\sin(A+B)} = \frac{a^2-b^2}{a^2+b^2}$$

or, $\dfrac{\sin(A-B)}{\sin(A+B)} = \dfrac{(2R\sin A)^2 - (2R\sin B)^2}{(2R\sin A)^2 + (2R\sin B)^2}$

or, $\dfrac{\sin(A-B)}{\sin(A+B)} = \dfrac{\cancel{4R^2}(\sin^2 A - \sin^2 B)}{\cancel{4R^2}(\sin^2 A + \sin^2 B)}$

or, $\dfrac{\sin(A-B)}{\sin(A+B)} = \dfrac{\sin(A+B)\cdot\sin(A-B)}{\sin^2 A + \sin^2 B}$

or, $\dfrac{\sin(A-B)}{\sin C} = \dfrac{\sin C \cdot \sin(A-B)}{\sin^2 A + \sin^2 B}$

or, $\sin(A-B)(\sin^2 A + \sin^2 B) = \sin^2 C \sin(A-B)$

or, $\sin(A-B)[\sin^2 A + \sin^2 B - \sin^2 C] = 0$

Either,

$\sin(A-B) = 0$

$A - B = 0$

$A = B$ [isosceles)

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or, $\sin^2 A + \sin^2 B - \sin^2 C = 0$

or, $\left(\frac{a}{2R}\right)^2 + \left(\frac{b}{2R}\right)^2 - \left(\frac{c}{2R}\right)^2 = 0$

or, $a^2 + b^2 - c^2 = 0$

or, $a^2 + b^2 = c^2$ [right angled] proved.

5. In $\triangle PQR$, if $(p+q+r)(p-q-r) + 3qr = 0$, find $\angle P$.

→ Given,

$(p+q+r)(p-q-r) + 3qr = 0$

or, $p^2 - \cancel{pq} - \cancel{pr} + \cancel{pq} - q^2 - qr + \cancel{pr} - qr - r^2 + 3qr = 0$

or, $p^2 - q^2 - r^2 - 2qr + 3qr = 0$

or, $p^2 - q^2 - r^2 + qr = 0$

or, $q^2 + r^2 - p^2 = qr$

or, $\frac{q^2 + r^2 - p^2}{2qr} = \frac{\cancel{qr}}{2\cancel{qr}}$

or, $\cos P = \frac{1}{2}$

$P = \cos^{-1}(1/2)$

$\therefore P = 60°$

6. In $\triangle ABC$, if $\frac{1}{a+c} + \frac{1}{b+c} = \frac{3}{a+b+c}$, then prove that $\angle C = 60°$.

→ Given,

$\frac{1}{a+c} + \frac{1}{b+c} = \frac{3}{a+b+c}$

or, $\frac{b+c+a+c}{(a+c)(b+c)} = \frac{3}{a+b+c}$

or, $(a+b+2c)(a+b+c) = 3(ab+ac+bc+c^2)$

or, $a^2 + ab + ac + ab + b^2 + bc + 2ac + 2bc + 2c^2$
$= 3ab + 3ac + 3bc + 3c^2$

or, $a^2 + b^2 + 2c^2 - 3c^2 + 2ab - 3ab + 3\cancel{ac} - 3\cancel{ac}$
$+ 3\cancel{bc} - 3\cancel{bc} = 0$

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or, $a^2+b^2-c^2 = ab$

or, $\dfrac{a^2+b^2-c^2}{2ab} = \dfrac{ab}{2ab}$

or, $\cos C = \dfrac{1}{2}$

or, $C = \cos^{-1}(1/2)$

$\therefore C = 60^\circ$

9. In $\triangle PQR$, if $\dfrac{\sin P}{\sin R} = \dfrac{\sin(P-Q)}{\sin(Q-R)}$, prove that p^2, q^2, r^2 are in A.P.

→ Given,

$\dfrac{\sin P}{\sin R} = \dfrac{\sin(P-Q)}{\sin(Q-R)}$

or, $\dfrac{\sin(Q+R)}{\sin(P+Q)} = \dfrac{\sin(P-Q)}{\sin(Q-R)}$

or, $\sin(Q+R)\cdot\sin(Q-R) = \sin(P+Q)\cdot\sin(P-Q)$

or, $\sin^2 Q - \sin^2 R = \sin^2 P - \sin^2 Q$

or, $\sin^2 Q + \sin^2 Q = \sin^2 P + \sin^2 R$

or, $2\sin^2 Q = \sin^2 P + \sin^2 R$

or, $2 \times \dfrac{q^2}{4R^2} = \dfrac{p^2}{4R^2} + \dfrac{r^2}{4R^2}$

or, $2 \times \dfrac{q^2}{\cancel{4R^2}} = \dfrac{1}{\cancel{4R^2}}(p^2+r^2)$

or, $q^2 = \dfrac{p^2+r^2}{2}$

$\therefore p^2, q^2, r^2$ are in A.P.

11. If the cosines of two of the angles of a triangle are proportional to the opposite sides, prove that the triangle is isosceles.

↳ Soln,

Let ABC be a triangle such that,

$$\frac{\cos A}{a} = \frac{\cos B}{b}$$

or, $\dfrac{\cos A}{\cancel{2R}\sin A} = \dfrac{\cos B}{\cancel{2R}\sin B}$

or, $\cot A = \cot B$

$A = B$

$\therefore \triangle ABC$ is an isosceles $\triangle$

proved

12. In $\triangle XYZ$, if $x\cos X = y\cos Y$ show that the triangle is either isosceles or right angled.

↳ Soln,

$$x\cos X = y\cos Y$$

or, $\dfrac{\cos X}{y} = \dfrac{\cos Y}{x}$

or, $\dfrac{\cos X}{\cancel{2R}\sin Y} = \dfrac{\cos Y}{\cancel{2R}\sin X}$

or, $\sin X.\cos X - \cos Y.\sin Y = 0$

or, $\dfrac{1}{2}(\sin 2X - \sin 2Y) = 0$

or, $\cancel{2}\sin\left(\dfrac{2x-2y}{2}\right)\cos\left(\dfrac{2x+2y}{2}\right) = 0$

~~or, sin2x = sin2y ⇒ x = y (Isosceles)~~

Either,

$\sin\left(\dfrac{x-y}{2}\right) = \sin 0$ $\qquad$ $\cos\left(\dfrac{2(x+y)}{2}\right) = \cos 90$

$x - y = 0$ $\qquad$ $x + y = 90°$ (Right angled)

$x = y$ (Isosceles)

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13. If $b-a=mc$, prove that $\cot\left(\frac{B-A}{2}\right)$

$$= \frac{1+m\cos B}{m\sin B}$$

→ R.H.S,

$$\frac{1+m\cos B}{m\sin B} = \frac{1+\left(\frac{b-a}{c}\right)\cos B}{\frac{b-a}{c}\sin B}$$

$$= \frac{c+(b-a)\cos B}{(b-a)\sin B}$$

$$= \frac{\cancel{a\cos B}+b\cos A+b\cos B-\cancel{a\cos B}}{(b-a)\sin B}$$

$$= \frac{b(\cos A+\cos B)}{(b-a)\sin B}$$

$$= \frac{2R\sin B(\cos A+\cos B)}{(2R\sin B-2R\sin A)\sin B}$$

$$= \frac{\cos A+\cos B}{\sin B-\sin A}$$

$$= \frac{\cancel{2}\cancel{\cos\left(\frac{B+A}{2}\right)}\cdot\cos\left(\frac{B-A}{2}\right)}{\cancel{2}\cancel{\cos\left(\frac{B+A}{2}\right)}\cdot\sin\left(\frac{B-A}{2}\right)}$$

$$= \cot\left(\frac{B-A}{2}\right)$$

= L.H.S

Proved.

14) In ΔABC, if $(\cos A + 2\cos C) : (\cos A + 2\cos B) = \sin B : \sin C$, prove that the triangle is either isosceles or right angle.

→ Given,

$$\frac{\cos A + 2\cos C}{\cos A + 2\cos B} = \frac{\sin B}{\sin C}$$

or, $\cos A \cdot \sin C + 2\cos C \cdot \sin C = \sin B \cdot \cos A + 2\sin B \cdot \cos B$

or, $\cos A \cdot \sin C + \sin 2C = \cos A \cdot \sin B + \sin 2B$

or, $\cos A \cdot \sin C - \cos A \cdot \sin B = \sin 2B - \sin 2C$

or, $\cos A(\sin C - \sin B) + \sin 2C - \sin 2B = 0$

or, $\cos A(\sin C - \sin B) + 2\cos(C+B) \cdot \sin(C-B) = 0$

or, $\cos A(\sin C - \sin B) + 2\cos(\pi - A) \cdot \sin(C-B) = 0$

or, $\cos A(\sin C - \sin B) - 2\cos A \cdot \sin(C-B) = 0$

or, $\cos A[\sin C - \sin B - 2\sin(C-B)] = 0$

or, $\cos A\left[2\cos\left(\frac{C+B}{2}\right) \cdot \sin\left(\frac{C-B}{2}\right) - 2\sin(C-B)\right] = 0$

or, $2\cos A\left[\cos\left(\frac{C+B}{2}\right) \cdot \sin\left(\frac{C-B}{2}\right) - 2\sin\left(\frac{C-B}{2}\right) \cdot \cos\left(\frac{C-B}{2}\right)\right] = 0$

or, $2\cos A \cdot \sin\left(\frac{C-B}{2}\right)\left[\cos\left(\frac{C+B}{2}\right) - 2\cos\left(\frac{C-B}{2}\right)\right] = 0$

As $\left[\cos\left(\frac{C+B}{2}\right) - 2\cos\left(\frac{C-B}{2}\right)\right] \neq 0$

So,

Either, $\cos A = 0$

$\cos A = \cos 90^\circ$

$\Rightarrow A = 90^\circ$

$\therefore \Delta ABC$ is right angled.

Or,

$\sin\left(\frac{C-B}{2}\right) = 0$

or, $\frac{C-B}{2} = 0$

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or, $C-B=0$

$\therefore C=B$

$\Rightarrow \Delta ABC$ is an isosceles Δ.

$\therefore \Delta ABC$ is either isosceles or Right angled #

15. In any ΔABC, if $\frac{\sin A+\cos A}{\cos B}=\sqrt{2}$, Show that $\angle C=135°$.

↳ Soln,

$$\frac{\sin A+\cos A}{\cos B}=\sqrt{2}$$

or, $\frac{\sin A+\cos A}{\sqrt{2}}=\cos B$

or, $\frac{1}{\sqrt{2}}\sin A+\frac{1}{\sqrt{2}}\cos A=\cos B$

or, $\sin 45°.\sin A+\cos 45.\cos A=\cos B$

or, $\cos(A-45)=\cos B$

or, $\cos(A-45)=\cos(-B)$

$A-45=-B$

$A+B=45°$

Now,

$A+B+C=180°$

$45+C=180°$

$\angle C=180°-45°$

$\therefore \angle C=135°$ #.